

Rolls-Royce Pegasus

Designed and built by Bristol Siddeley (which became Rolls-Royce Bristol), this vertical/short takeoff and landing (V/STOL) powerplant is an audacious and technically brilliant piece of engineering. Its elegant design incorporates four swiveling thrust nozzles instead of the conventional single, rearward-facing one. It powered the iconic Harrier jump jet to world-renowned success.

AN IDEA WITH LEGS

The Pegasus engine's four thrust nozzles give a lightly loaded Harrier the maneuverability of a helicopter. With the four stable "legs" of thrust on which to stand, the pilot can then progressively rotate the thrust line to make the transition to conventional forward flight. The vertical or short takeoff and landing capabilities eliminate the need for conventional runways and offer a major advantage at sea, where Harriers operate from a wide variety of ships.

Rear (hot) nozzle
This is rotatable through 98.5 degrees to allow transition from vertical to horizontal flight.

Nozzle profile
This is curved to maximize thrust.

Nozzle vane

Cooling air manifold
prevents nozzle rotation bearing seizing.

Insulation blanket
This protects the airframe from the extreme heat of engine exhaust gases.

Port (hot) nozzle

Water-injection delivery pipe
Water is sprayed into the turbine for takeoff to stop the blades overheating.

Turbine casing

Fire extinguisher feed pipe

Collar seal

Diffuser casing

Stiffening ribs on casing
This improves rigidity and vibration-proofing without adding weight.

